

Foundations of Cybersecurity/Applied Cryptography

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EXERCISE N. 1 [12]

With reference to multiple encryption

1. Illustrate the cases in which it is appropriate to use it.
2. Describe the meet-in-the-middle attack, arguing its complexity.
3. Describe 3DES-EDE and discuss its properties.

EXERCISE N. 2 [10] [ALL]

The following identification protocol allows a mobile station (MS) to identify itself w.r.t. an access

M1 MS → AP: REQ

M2 AP → MS: chl

M3 MS → AP: iv, rsp

M4 AP → MS: **true|false**

point (AP). In the protocol, chl denotes a 128-bit random challenge; rsp the corresponding response; and iv a 24-bit random initialization vector.

Upon receiving message M2, MS computes rsp as $rsp = \text{PRBG}(k||iv) \oplus chl$ where $\text{PRBG}(k||iv)$ is a 128-bit sequence generated by a secure pseudo-random generator

PRBG. The generator is seeded by $k||iv$, where k is a long-term cryptographic key secretly shared by AP and MS and established in off-line fashion.

Upon receiving message M3, AP computes $rsp' = \text{PRBG}(k||iv) \oplus chl$ and returns ($rsp == rsp'$) to MS.

1. Does this protocol guarantee identification? Is a passive adversary able to impersonate a mobile station? If so, discuss the complexity of the attack.
2. Let us suppose now that AP sends MS the initialization vector iv together with the challenge chl in message M2 which becomes $M2 = \langle iv, chl \rangle$. Field iv in M3 is not necessary anymore. Is a passive adversary able to impersonate a mobile station? If so, discuss the complexity of the attack

EXERCISE N. 3 [8]

Tell whether the following statements are true or not. Argue the answer.

1. A Vernan cipher is unconditionally secure provided its key is at least 64 bits.
2. In a cipher E, the number of keys is not smaller than the number of plaintext messages. Then E is perfect.
3. Let E be a block cipher with $k = 128$ and $n = 128$, then E can be unconditionally secure.
4. An unconditionally secure cipher is also computationally secure.

SOLUTION

EXERCISE N.1

See theory.

EXERCISE N.2

- 1 An adversary can eavesdrop the channel, and obtains iv , rsp and chl . So, it can determine $z = rsp \oplus chl$, where $z = \text{SPRG}(k||iv)$. By using the pair (iv, z) then the adversary can identify as MS w.r.t. to AP as many times he likes.
- 2 As in step A, the adversary can eavesdrop the channel, and obtains iv , rsp and chl . However, now, the choice of iv is not under his/her control. So (s)he must build a dictionary (iv, z) . Whenever AP reuses iv , the adversary can reuse z . Let E denote the number of entries of the dictionary. Then $E = 2^{24}$. Each entry must accommodate the pseudo-random sequence z of bits corresponding to iv . Let $Z = \text{sizeof}(z)$, then $Z = 16$ (in bytes). It follows that the dictionary size D is $D = E \times Z = 2^{28}$ bytes = 256 Mbytes.

EXERCISE N.3

Tell whether the following statements are true or not. Argue the answer.

1. A Vernan cipher is unconditionally secure provided its key is at least 64 bits. FALSE. Vernam cipher is unconditionally secure provided the key is perfectly random.
2. In a cipher E , the number of keys is not smaller than the number of plaintext messages. Then E is perfect. FALSE. The condition is necessary, not sufficient.
3. Let E be a block cipher with $k = 128$ and $n = 128$, then E can be unconditionally secure. FALSE. For the block cipher to be perfect, the block cipher should implement $2^{128}!$ permutations. Actually, the block cipher E implements $2^{128} \ll 2^{128}!$.
4. An unconditionally secure cipher is also computationally secure. TRUE.